Mark schemes

Q1.

(a) H_0: $\mu = 175$

H_1: $\mu < 175$

Both; accept H0: $\mu \geq 175$
Do not accept mean or $\overline{x}$
but accept population mean

B1

$\overline{x} = 168.1$

B1

$$z = \frac{\text{'}168.1\text{'} - 175}{9.4 / \sqrt{6}}$$

For use of 9.4 / √6

M1

For rest of formula (ignore sign)

m1

$= -1.798$

Must be negative AWRT −1.80

A1

CV = − 1.6449

AWFW − 1.64 to − 1.65

B1

−1.6449 > −1.798 so test statistic in critical region

Comparison of correct test statistic with correct CV
***must be seen** (diagram or words)*

Reject H_0, significant evidence that batch **mean** is less than 175 grams

OE; suspicion supported
Must be in context AG

A1

7

(b) H_0: $\mu = 175$

H_1: $\mu < 175$

*Award B1 for both correct if not scored in **(a)***

$t = \dfrac{169.4 - 175}{11.2 / \sqrt{20}}$

For use of 11.2 / √20

M1

For rest of formula (ignore sign)

m1

= − 2.236

Must be negative AWRT − 2.24

A1

CV(t_{19}) = − 2.539

AWRT − 2.54

B1

− 2.236 > − 2.539 so test statistic not in critical region

Comparison of correct test statistic with correct CV (need not be seen)

Accept H_0, no significant evidence that batch mean / weight is less than 175 grams

OE; suspicion not supported

A1

5

(c) Because the significance level is 1% instead of 5%

OE; eg SL is different

Reference to sample size ⇒ E0

E1

1

[13]

Q2.

(a) $H_0 : \mu = 65$

B1

$H_1 : \mu > 65$

B1

2

(b) $\bar{X} \sim N\left(\mu, \frac{\sigma^2}{n}\right)$ = N (65, 0.64)

for 0.64

B1

for Normal and 65

B1

2

(c) P(Type I error) =
P(H_0 rejected when H_0 true)
= P($\bar{X}$>66.4 when μ = 65)
= P$\left(Z > \frac{66.4 - 65}{0.8}\right)$

M1

= P(Z > 1.75)

area change

m1

= 1 – 0.95994

= 0.04006

AWRT 0.040

A1

∴ significance level of test ≈ 4%

ft on Type I error

A1ft

2

4

(d) P (Type II error) =

P (H_0 accepted when H_0 false)

P ($\bar{X}$< 66.4 when μ = 67)

= P$\left(Z < \frac{66.4 - 67}{0.8}\right)$

M1

= P(Z < –0.75)

A1

= 1 – Φ (0.75)
= 1 – 0.77337

m1

= 0.22663
= 0.227

AWRT 0.23

A1

4

[12]

Q3.

(a) $\frac{1}{4}$, $\frac{1}{4}$ inserted in table

Accept statement, or use of correct probs in (b)

B1

1

(b) Method for mean

Award M1 A0 if c then divides, e.g. by 6

M1

Mean = 4

Allow NMS

A1

E(X^2) = 19

PI; allow even if c gives this as variance.

B1

Variance = $19 - 4^2 = 3$

Alternative: $\Sigma p(x - 4)^2 = 3$ M1 A1 (–1 EE); Allow NMS

M1 A1

[6]

Q4.

(a) $H_0 : \mu = 500$
$H_1 : \mu < 500$

B1

1

(b) $\bar{Y} \sim N(500, 0.8^2)$

For Normal and 500

B1

For 0.64

B1

2

(c) (i) P(Type I error) = 0.05

B1

1

(ii) $\frac{\bar{y} - 500}{0.8} = -1.6449$

Correct z-value (Accept ± 1.645 or ±1.64)

B1

$\bar{y} = 498.68$

M1

$\therefore$ Critical region for $\bar{y}$ is

$\bar{Y} < 498.68$

(AWFW 498.68 and 498.69) (must be a region)

A1

3

(d) P(Type II error) = P($\bar{Y}$ > 498.68)

Can be $\Rightarrow$ by what follows or diagram.

M1

$= P\left(Z > \frac{4.98.68 - 497.5}{0.8}\right)$

for use of N(4.97.5, 0.8^2*)*

for $\frac{\bar{y} - 497.5}{0.8}$

M1A1f.t.

= P(Z > 1.4801)

(AWFW 1.48 to 1.49)

A1f.t.

= 1 – Φ(1.48)

for 1 – $\Phi(z)$

M1

= 1 – 0.93056

(AWRT 0.931)

B1

= 0.0694

AWFW 0.0680 to 0.0702

A1

7

[14]

Q5.

(a) (i) $\overline{X} \sim N\left(40, \frac{0.64}{50}\right)$ or $N\left(40, \frac{0.8^2}{50}\right)$

or N (40, 0.0128)

for Normal & 40

Idea of $\frac{\sigma^2}{n}$*, correct* $\frac{\sigma^2}{n}$ *seen M1*

B1/M1A1

(ii) P(Type I error) = 0.05

B1

4

(b) Acceptance region

for [–1.96, 1.96]{wrong symmetrical M1A0}

M1A1

= (40 – 1.96 × 0.113, 40 + 1.96 × 0.0113)

$\frac{\sigma}{\sqrt{n}} = 0.1131$ (A1)

M1A1

= (39.78, 40.22)

(On their z)

A1f.t.

5

(c) (i) Accepting H_0 when H_0 incorrect

B1

(ii) $P(\overline{X} < 39.78) = P\left(Z < \frac{39.78 - 40.3}{0.1131}\right)$

= P(Z < – 4.61)

ft on their z from (b)
(allow – 4.60)

M1A1f.t.

$$P(\overline{X} < 40.22) = P\left(Z < \frac{40.22 - 40.3}{0.1131}\right)$$

$= P(Z < – 0.6917)$
(allow – 0.7073)

A1f.t.

$\therefore$ P(Type II error) = 1 – $\Phi\lambda$(0.69) – $\Phi\lambda$(–4.61)
= (1 – 0.75490) – 0
= 0.2451
(allow 0.24 to 0.25)
(allow 1– 0.76115)

M1

(allow 0.239)

A1

6

AWRT

[15]

Q6.

(a) (i) $\bar{x} = 46.5 \quad s = 10.81$

46.5 cao and 10.8 (10.8 to 10.82)

B1

H_0: μ = 40

B1

H_1: $\mu \neq 40$
one correct hypothesis – generous
both hypotheses correct – ungenerous
Allow H_1 $\mu > 40$

B1

$$t = \frac{(46.5 - 40)}{\frac{10.81}{\sqrt{8}}} = 1.70$$

use of their $\frac{\text{s.d.}}{\sqrt{8}}$

M1

completely correct method for t ignore sign

m1

1.70 (1.695 to 1.705)

A1

c.v t_7 ± 2.365; 1.70 lies between

7df

B1

± 2.365 so accept H_0 , mean is 40 mins

2.365 – their df – ignore sign
(cv 1.895 for one tail test)

B1ft

correct conclusion – must be compared with t

A1ft

Alternatively Allow confidence interval approach

(ii) $H_0 : \mu = 50$
$H_1 : \mu \neq 50$

both hypotheses correct – ungenerous Allow H_1 $\mu < 50$

B1

$$t = \frac{(46.5 - 50)}{\frac{10.81}{\sqrt{8}}} = 0.916$$

–0.916 (–0.915 to –0.916)

A1

c.v t_7 ± 2.365;–0.916 lies between

(cv 1.895 for one tail test)

±2.365 so accept H_0 , mean is 50 mins

correct conclusion must be compared with
both tails or lower tail of t

A1ft

12

Alternatively Allow confidence interval approach

N.B. Mark a(ii) as a(i) and a(i) as a(ii) if more favourable

(b) Claim 1. **C** Not true – no null hypothesis rejected so no Type 1 error made

correct conclusion for correct reason – be generous for E1 but disallow no or clearly incorrect reason

E2,1

Claim 2. **B** Possibly true – true if population mean is equal to neither 40 nor 50

correct conclusion for correct reason – be generous for E1 but disallow no or clearly incorrect reason

E2,1

Claim 3. **A** Definitely true – since mean cannot equal both 40 and 50

correct conclusion for correct reason – be generous for E1 but disallow no or clearly incorrect reason

E2,1

6

[18]

Q7.

(a) $\bar{x} = 37.75$ $s = 4.6928$

37.75(37.7, 37.8) & 4.69 (4.69, 4.7)

B1

H_0: $\mu = 40$

One correct hypothesis - generous

B1

H_1: $\mu < 40$

Both hypotheses correct - ungenerous

B1

$t = \dfrac{37.75}{\dfrac{4.6928}{\sqrt{12}}} = -1.66$

use of candidate $\dfrac{s}{\sqrt{12}}$

method for t- ignore sign

Completely correct method for t

–1.66 (–1.65, –1.67)

M1 m1 m1 A1

c.v. t_{11} – 1.796 accept H_0, conclude no significant evidence to show mean is less than 40 months

11 df

–1.796 (–1.79, –1.8) ignore sign

Conclusion, must be compared with correct tail *of t-distribution*

B1 B1 A1

10

(b) H_0: $\mu = 40$

H_1: $\mu < 40$

Both hypotheses correct

B1

$$z = \frac{39.2 - 40}{\frac{4.2}{\sqrt{160}}} = -2.41$$

Method for z

M1

c.v. –1.6449 reject H_0, significant evidence to show mean is less than 40.

–2.41 (–2.4, –2.42) and

–1.6449(–1.64, –1.655) ignore sign

Conclusion, must be compared with correct tail of z or t.

A1 A1ft

4

(c) (i) neither, both 5%

neither

both 5%

B1 E1

(ii) neither - cannot make a Type II error if mean is 40

neither

No chance of type II error

B1 E1

(iii) (a), smaller sample

(a)

Smaller sample

B1 E1

6

[20]

Q8.

(a) $H_0 : \mu = 200$

$H_1 : \mu > 200$

$\bar{x} = 205.545$

one correct hypothesis – generous

B1

both hypotheses correct – ungenerous

B1

$$z = \frac{205.545 - 200}{\frac{6}{\sqrt{11}}} = 3.07$$

use of $\frac{6}{\sqrt{11}}$

M1

correct method for z - ignore sign

m1

3.07 (3.06 to 3.07)

A1

cv for 5% 1-sided risk 1.6449

1.6449 (1.64 to 1.65)

B1

Reject H_0

Correct conclusion - must be compared with correct tail of z

A1f.t.

7

Conclude mean time to spoilage is greater than 200 hours.

(b) No problem unless the 3 containers sold could not be treated as a random sample.

Comment on randomness of 3 containers

E1

Unlikely to bias sample.

Correct deduction
Allow comment on reduced sample size

E1

2

(c) Claiming mean time to spoilage is more than 200 hours when it isn't

Idea of Type 1 error

E1

In context - must be 1-sided

E1

2

(d) (i) 0.05

0.05 ***cao***

B1

(ii) 0

0 ***cao***

B2

3

[14]

Examiner reports

Q1.

The hypotheses were generally well expressed in terms of μ although often ≠ was used in H_1. Although the sign of the test statistic was usually correct, there was much confusion over the sign of the critical value. Sometimes a negative test statistic was compared with a positive critical value (often with a diagram showing that this was exactly what was being done). Frequently the critical value was stated as ±1.6449 (sometimes with two-tailed diagrams) and from the two alternatives the negative was chosen because the test statistic was negative. The implication was that a test statistic of +1.798 would have been compared with +1.6449 and also resulted in the rejection of H_0 (as seemed to be the case where a modulus sign was used). The use of t_{19} rather than z was good in part (b). In part (c), most students appreciated the effect of the significance level in setting the probability of a Type I error.

Q2.

The hypotheses were usually stated correctly, although there were a few candidates who used $\bar{x}$ instead of μ or who missed out an attempt at μ. altogether. Part (b) was surprisingly poorly done, with many candidates stating N(65, 32) but then going on to use the correct distribution for $\bar{x}$ in part (c). Many candidates calculated correctly the probability of a Type I error in part (c), but then either failed to give the significance level of the test or simply quoted it incorrectly as 5%. Part (d) was answered better than in previous series but it continued to pose problems for average candidates.

Q3.

This question was very well answered, with many candidates gaining full marks.

A common error in finding the mean was to divide the correct answer by 6. In calculating the variance, a substantial minority of candidates squared the probabilities instead of, or as well as, the values of the random variable. Another common mistake was to subtract the mean, rather than the square of the mean.

Q4.

In part (a) the null hypothesis, $H_0 : \mu = 500$, and the alternative hypothesis $H_1 : \mu < 500$ were often correctly stated. Most candidates also correctly quoted $\bar{Y}$ ~ N(500, 0.64) in part (b) and then used σ = 0.8 in the remainder of the question.

In part (c)(i) most candidates answered 5% with some converting this to 0.05. Some candidates thought that 5% is equivalent to 0.5. Many candidates found $\bar{Y}$ = 498.68 in part (c)(ii) but did not go on to determine the critical region. Candidates were expected to state $\bar{Y}$ < 498.68. Some candidates considered this as a 2-tail test using z = ±1.95 whilst others thought that z = ±1.65was an acceptable approximation for z = –1.6449.

Candidates performed well when dealing with the calculation of a Type II error in part (d). Many good solutions were seen, but some candidates made the question more difficult by wrongly using a 2-tail configuration.

Q5.

Candidates found this to be the most difficult question on the paper. In part (a)(i), $\overline{X} \sim$ N(40, 0.0128) or its equivalent was often seen but many candidates thought that $\overline{X} \sim$ N$\left(40, \frac{0.8}{\sqrt{50}}\right)$, even though σ = $\frac{0.8}{\sqrt{50}}$ was correctly used in later calculations. Part (a)(ii) was often answered very well with 5% being the most popular answer seen. Some candidates unfortunately thought that this equated to 0.5, rather than the correct response of 0.05, whilst others simply described what they thought was a Type I error without giving a value for P (Type I error), as requested. This latter response gained no credit.

In part (b), most candidates realised from looking at the given hypotheses that a 2-tailed test was being considered. Many candidates used a diagram with z = ±1.96 to demonstrate the region required, but then failed to continue the process and consider 40 ± 1.96 $\frac{\sigma}{\sqrt{n}}$ to obtain [39.78, 40.22] as the acceptance region asked for in the question. Some candidates who followed the correct method throughout lost credit for not writing the answer correct to two decimal places.

Part (c)(i) produced mixed and often confusing answers from candidates. 'Accepting H_0 when H_0 incorrect' was the expected response here. Part (c)(ii) produced some excellent solutions but only from the most able candidates. The vast majority of candidates seemed to have very little or no idea of what was expected of them. Although this part of the question covered probably the most difficult concept in the unit, it was disappointing to see the lack of understanding apparent in candidates. Candidates who used some type of diagrammatic representation when finding the probability of a Type II error usually gained some credit for their efforts.

Q6.

This question was extremely well answered. In particular, in part (a)(ii), the majority of candidates correctly calculated a negative test statistic and compared it with a negative critical value. Part (b) was expected to offer a substantial challenge. However many candidates obtained full marks.

Q7.

Parts (a) and (b) were answered well. In particular, the great majority of answers showed correct signs for both test statistics and critical values. Part (c) was intended to refer to the nature of the tests. However, many candidates referred to the particular application and this was accepted. For example, in part (c)(i) the expected answer was "neither, since both tests have a 5% risk of Type 1 error" but answers such as "(b) - a Type 1 error cannot have been made in (a) because H_0 was accepted" were allowed.

Q8.

Candidates were well prepared for parts (a) and (c). Most made clear in words or, more usually, symbols that their hypotheses related to a population. Confusion between one-sided and two-sided tests was not common.

Some good efforts at part (b) were made with many saying that there would be no problem provided that the three tubs sold were randomly selected, and others commenting on the smaller sample size. Answers which implied that a random selection might lead to problems if the three longest lasting tubs of yoghurt were chosen received no credit, but those who gave a reason as to why the three longest lasting tubs might be chosen – e.g. the freshest looking tubs being chosen – received full marks.

Part (d) proved to be a good discriminator. The most common error amongst those who made a reasonable attempt was to answer “0” for part (i) and “0.05” for part (ii), instead of the other way round. Another popular answer, for both parts, was “0.95.”